



Project Plan

Asma Abbas

Topic and Data



- My topic has to do with educational attainment and poverty, within Maryland. I want to study the relationship between these two variables. For this project, I will be using three different datasets, all of them have generally equal importance to answer questions for this project.

Dataset 1: Sourced from Maryland Open Data. It has information regarding socioeconomic factors in Montgomery County. The key variables for this dataset are:

- Poverty Rate
- Income
- Unemployment
- Education levels

Dataset 2: Sourced from Census.Gov. This dataset contains similar information, but for counties all across Maryland. Key variables are:

- Poverty Rate
- Child poverty rate
- Median household income
- Education levels

Dataset 3: Sourced from the MCPS website. It has information directly from high schools in Montgomery County.

Key Variables:

- Graduation rates
- Dropout rate
- Attendance rate
- FARM participation



Questions

- How strongly is poverty linked to graduation rates?
- How does this relationship compare across different counties in Maryland?
- Does race/ethnicity affect the relationship?
- Are there geographic areas with stronger effects?
- Can income predict graduation outcomes?
- How does FARM participation relate to graduation rates?
- Do dropout rates correlate with poverty?
- How does attendance impact educational outcomes?

Methods and Analysis



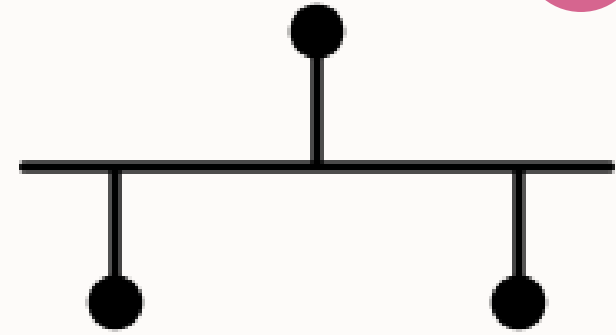
- The program I will be using for this project is R studio.
- To complete this project, I will be doing things like:
 - Data cleaning and filtering
 - Integration of datasets
 - EDA
 - Regression models
 - Visualizations: bar graphs, scatterplots, heatmaps
- Correlation Analysis
 - Find relationships
- Regression Models
 - Predict graduation rates
- Comparative analysis
 - County-level comparisons
- Spatial Analysis
 - Find geographic patterns

Risks and challenges



- I've never done a project of this scale before, so that will certainly be a challenge!
- Though the datasets have similar contents, there is a risk of misalignment. The schools might not match up with the areas that are mentioned in the census dataset.
- The Montgomery County dataset has a very limited time range
- I am also worried about weak correlations, but that remains to be seen.
- To work around these problems , I will
 - Try to aggregate geographic matches
 - Clean the data thoroughly
 - Focus on cross sectional analysis

Milestones and Timeline



- Week 1: Data cleaning
- Week 2: EDA and visualizations
- Week 3: Statistical analysis
- Week 4: Larger and comparative analysis
- Week 5: Final touches and presentation!

Goals:

With this project, I hope to thoroughly understand the relationship between poverty and education within our state. This is a right, and it's important to observe how poverty can impact a student. It's also important for Maryland to be an educated state, and gathering insights from this project will allow me to propose solutions. I will also be able to notice high risk areas, which should be targeted first when it comes to policy change.

Thank You!

